

Resolution Geometry of Observable Duration

Quotient Factorization, Admissible Process Geometry, Universal Clock Representations, and the Conditional Placement of General Relativity

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Abstract

Resolution Geometry (RG) begins with a declared statistical experiment and identifies latent states that induce the same complete observation law. The resulting observational quotient is canonical relative to the experiment: every experiment-internal prediction factors uniquely through it. This paper develops the temporal consequences of that elementary universal property without promoting time, spacetime, or any particular metric to primitive status. An admissibility protocol selects a maximal stable observable sector, supports natural transport, and supplies the domain on which observable processes may be composed. On the continuous positive commuting semigroup stratum, multiplicative process composition has a unique normalized continuous additive linearization, namely $\mathcal{T} = -\log C$ up to positive scale. Compatible scalar clock mechanisms form positive affine charts and glue to a common affine clock object precisely when their transition maps satisfy identity and cocycle conditions. For nonadditive multitime processes, the defect $\mathcal{M}_{012} = \mathcal{T}_{02} - \mathcal{T}_{01} - \mathcal{T}_{12}$ is identified as the local obstruction to strict duration gluing; its promotion to a global cohomology class is isolated as an open theorem rather than assumed.

We then place general relativity as a conditional realization of the same observable structure. The admissible sector may supply clock-path data and conserved source observables; together with a field-equation model, gauge quotient, initial or boundary data, sufficient path rank, stable Lorentzian signature, and held-out validation, these data may identify an observable metric class and its proper-time functional. This bridge is positive but non-faithful: stress-energy alone does not determine geometry, clock paths do not determine a metric without sufficient rank, unsigned proper duration does not determine time orientation, and successful reconstruction of a four-dimensional Lorentzian observable quotient does not identify a unique latent dimension or primitive temporal ontology. The resulting theorem ladder separates proved quotient, semigroup, affine-gluing, and no-go results from two remaining closures: a representability theorem for universal duration and an empirical cross-clock, multi-path identification with relativistic proper time.

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1 Scope and organizing claim

RG is not introduced as a derivation of general relativity, quantum mechanics, or information geometry. Its stable claim is narrower and more structural:

A declared experiment determines what can be distinguished; every prediction available to that experiment factors through the corresponding observable quotient; further geometric and physical structures are admissible enrichments or realizations of that quotient.

The hierarchy studied here is

$$(\mathcal{E}, \Gamma) \longrightarrow Q_{\mathcal{E}} \longrightarrow W_{\text{obs|adm}} \longrightarrow \text{transport} \longrightarrow C \longrightarrow \mathcal{T} \longrightarrow \text{clock representation} \longrightarrow \text{physical realization}.$$

No arrow is silently reversed. In particular, operational reconstruction of proper time does not imply that a primitive temporal dimension was required by the experiment.

2 Declared experiments and the canonical quotient

Definition 2.1 (Declared experiment). A declared experiment is $\mathcal{E} = (\Theta, \mathcal{O}, \{P_{\theta}\}_{\theta \in \Theta})$, where P_{θ} is the complete declared observation law. Define

$$\theta \sim_{\mathcal{E}} \theta' \iff P_{\theta} = P_{\theta'}$$

and let $q : \Theta \rightarrow Q_{\mathcal{E}} = \Theta / \sim_{\mathcal{E}}$ be the quotient map.

Theorem 2.2 (Experiment-internal factorization). *If $f : \Theta \rightarrow Z$ is constant on equality-of-law classes, then there is a unique map $\tilde{f} : Q_{\mathcal{E}} \rightarrow Z$ such that $f = \tilde{f} \circ q$.*

Proof. Set $\tilde{f}([\theta]) = f(\theta)$. Constancy on equivalence classes gives well-definedness; surjectivity of q gives uniqueness. $\square$

Corollary 2.3 (No experiment-internal latent refinement). *A distinction between two representatives of the same class in $Q_{\mathcal{E}}$ is not an experiment-internal prediction of $\mathcal{E}$.*

Remark 2.4. Canonicity here is exact but relative: changing the declared observation law changes the quotient. This is sufficient for a universal property and does not require a protocol-free ontology.

3 The admissibility contract

Definition 3.1 (Admissibility protocol). An admissibility protocol Γ specifies the regularity and comparison conditions under which observable directions, sectors, transports, and reconstructions are retained. The present protocol requires, where applicable: quotient invariance, locally stable rank, bounded conditioning, monotonicity under declared coarse-graining, composable transport, and continuity.

Definition 3.2 (Maximal admissible observable sector). $W_{\text{obs|adm}}$ is the maximal protocol-admissible sector on which the declared observable comparisons and transports remain quotient-invariant, rank-stable, and uniformly controlled on the domain under study.

Proposition 3.3 (Componentwise admissibility minimality). *Relative to the target properties below, each corresponding condition is independently necessary:*

1. *quotient invariance for experiment-internality;*
2. *rank stability for locally constant sector type;*
3. *bounded conditioning for uniform inverse stability;*
4. *coarse-graining monotonicity for noncreation of distinguishability;*
5. *transport composability for path-independent gluing;*
6. *continuity for perturbative stability.*

Proof. Each removal admits a finite witness. Representative-dependent maps distinguish equal-law states if quotient invariance is removed. The family $\text{diag}(1, \varepsilon)$ crosses any fixed numerical rank threshold as $\varepsilon \rightarrow 0$. Its inverse amplifies perturbations proportionally to ε^{-1} . A declared forgetting map not required to contract can enlarge distance. Noncomposable transports give different results along direct and composite paths. A threshold functional jumping at the identity violates perturbative stability. These witnesses prove necessity for the paired target properties, not global initiality of Γ among all conceivable protocols. $\square$

Non-claim 3.4. The proposition is not yet an initial-object theorem for admissibility protocols. Such a theorem requires a category of protocols, explicit protocol morphisms, and a precise meaning of factorization through Γ .

4 Observable processes and logarithmic duration

Definition 4.1 (Observable coherence process). On a finite-dimensional regular sector $W_{\text{obs|adm}}$, an observable coherence process is a family $(C_t)_{t \geq 0}$ of strictly positive self-adjoint contractions. On the semigroup stratum,

$$C_0 = I, \quad C_{t+s} = C_t C_s, \quad 0 \prec C_t \preceq I,$$

with continuity in t .

Theorem 4.2 (Operator duration). *There is a unique positive semidefinite generator K such that $C_t = e^{-tK}$, and*

$$\boxed{\mathcal{T}_t = -\log C_t = tK}$$

is additive and natural under admissible conjugating transport.

Proof. Finite-dimensional semigroup theory gives $C_t = e^{tA}$. Self-adjoint contractivity gives $A = -K$ with $K \succeq 0$. Strict positivity permits the principal logarithm. Functional calculus gives naturality under similarity on the stated domain. $\square$

Theorem 4.3 (Scalar logarithmic characterization). *Let $h : (0, 1] \rightarrow [0, \infty)$ be continuous, satisfy $h(1) = 0$, and obey $h(xy) = h(x) + h(y)$. Then $h(x) = -a \log x$ for some $a \geq 0$; strict monotonicity gives $a > 0$.*

Proof. Set $f(u) = h(e^{-u})$. Then $f(u + v) = f(u) + f(v)$ on $[0, \infty)$. Continuity implies $f(u) = au$. $\square$

Non-claim 4.4. The canonical object is operator-valued. The theorem neither selects a unique scalar clock nor identifies $\mathcal{T}$ with relativistic proper time.

5 Scalar clocks as representations

Definition 5.1 (Duration character). A scalar clock representation is a continuous monotone map χ from an operator-duration object into an ordered additive scalar target, subject to the operational semantics declared for the clock.

For example,

$$\mathrm{tr} \mathcal{T}, \quad r^{-1} \mathrm{tr} \mathcal{T}, \quad \|\mathcal{T}\|, \quad \langle v, \mathcal{T}v \rangle$$

are generally distinct. They answer extensive, intensive, extremal, and modal questions respectively.

Theorem 5.2 (Extensive determinant character). *Let $\chi_n : \mathrm{PD}_n(\mathbb{C}) \rightarrow \mathbb{R}_{>0}$ be continuous, unitarily invariant, multiplicative on commuting positive operators, and extensive under direct sums. Then*

$$\chi_n(A) = (\det A)^a$$

for a dimension-independent $a \geq 0$. Consequently, the associated extensive scalar duration is

$$-\log \chi(C) = a \mathrm{tr}(-\log C) = a \mathrm{tr} \mathcal{T}.$$

Proof. Unitary invariance reduces $\log \chi$ to a symmetric spectral function. Direct-sum extensivity reduces it to a sum of one-variable terms. Multiplicativity gives the continuous logarithmic Cauchy equation in each eigenvalue. $\square$

Corollary 5.3. *Uniqueness of an extensive scalar clock is only up to positive unit. Other scalar semantics are not excluded because they do not satisfy the same characterization axioms.*

6 Affine clock gluing

Definition 6.1 (Clock atlas). A clock chart on an observable-history domain U_i is a nondegenerate reading map $R_i : U_i \rightarrow \mathbb{R}$. Charts are compatible when

$$R_j = a_{ji} R_i + b_{ji}, \quad a_{ji} > 0,$$

and on triple overlaps

$$a_{ki} = a_{kj} a_{ji}, \quad b_{ki} = a_{kj} b_{ji} + b_{kj}.$$

Theorem 6.2 (Universal affine clock factorization). *A connected compatible clock atlas glues to a one-dimensional affine clock object $\mathbb{T}$. Each R_i factors through a common coordinate τ on $\mathbb{T}$, and the construction is unique up to one global positive affine transformation. For elapsed duration the remaining gauge is positive scale.*

Proof. Choose one chart as reference and transport its affine coordinate through the overlap graph. The cocycle identities make the result path-independent. Any second choice differs on the reference chart by one positive affine map, which then propagates globally. $\square$

No-go theorem 6.3 (History-dependent recalibration obstruction). *If overlap comparison requires nonlinear or history-dependent recalibration, the mechanisms do not define charts of the same affine clock under the declared calibration category.*

7 Memory and the obstruction to global duration

Definition 7.1 (Local composition defect). For composable intervals define

$$\mathcal{M}_{012} = \mathcal{T}_{02} - \mathcal{T}_{01} - \mathcal{T}_{12}.$$

For ordered noncommuting exponentials this may include a BCH defect; operational memory requires a multitime factorization criterion and must not be identified with noncommutativity alone.

Proposition 7.2 (Local obstruction). *On a single compositional triangle, strict additive gluing holds if and only if $\mathcal{M}_{012} = 0$.*

Proof. This is immediate from the definition. $\square$

Conjecture 7.3 (Global obstruction theorem). Given a specified process nerve $\mathcal{N}$, coefficient duration object $\mathcal{A}$, and compatible gauge action, local defects define a cocycle $\mathcal{M}$. Under appropriate regularity and connectivity hypotheses,

$$[\mathcal{M}] = 0 \in H^2(\mathcal{N}; \mathcal{A}) \iff \text{local duration data admit a global additive potential.}$$

Non-claim 7.4. The finite triangle identity does not by itself prove the global cohomological statement.

8 Universal duration as a representability problem

Definition 8.1 (Admissible duration assignment). Let $\mathbf{ObsProc}_\Gamma$ be a category of quotient-identified admissible observable processes. An A -valued duration assignment is a quotient-invariant, natural, continuous map compatible with identity and composition in the declared additive or cocycle-corrected sense.

Conjecture 8.2 (Universal duration representability). There exists a functorial duration object $\mathfrak{T}$ in a category $\mathbf{Dur}$ such that, naturally in the additive target A ,

$$\mathrm{Hom}_{\mathbf{Dur}}(\mathfrak{T}(X), A) \cong \mathrm{Dur}_\Gamma(X; A).$$

Equivalently, every admissible duration assignment factors as

$$\Lambda = \chi_\Lambda \circ \mathfrak{T},$$

with uniqueness relative to the target category and normalization gauge.

Proposition 8.3 (Semigroup realization of the conjecture). *On the positive commuting scalar semigroup stratum, with normalization $h(e^{-1}) = 1$, the representing additive coordinate is $-\log C$.*

Proof. The scalar logarithmic characterization gives existence and uniqueness in the declared continuous normalized class. $\square$

Remark 8.4. Quotient invariance alone cannot prove uniqueness: $\mathcal{T}$ and $2\mathcal{T}$ satisfy the same structural axioms absent normalization. Operator-valued uniqueness also requires a separating or complete family of characters.

9 Why process tensors remain open

For a process tensor or quantum comb Υ , composition is generally a link product, not multiplication of positive scalar survival factors. A principal logarithm may fail to be a positive generator, may depend on branch or representation, and need not linearize the link product. A correct extension must either identify a divisible generator stratum or retain a cocycle-valued object $(\mathfrak{T}, [\mathcal{M}])$. This is the main mathematical frontier for universal duration.

10 The GR realization bridge

Definition 10.1 (GR realization data). A GR realization of an admissible observable sector consists of:

1. scalar clock/path observables obtained from $\mathfrak{T}$ or its semigroup representative;
2. observable source data proposed as $T_{\mu\nu}$;
3. a Lorentzian competitor class and gauge quotient;
4. initial or boundary data;
5. field equations and conservation constraints;
6. a sufficiently rich and stably conditioned path design.

The bridge is a joint constrained inverse problem:

$$\begin{cases} G_{\mu\nu}(g) = 8\pi T_{\mu\nu}, \\ \nabla_{\mu}^{(g)} T^{\mu\nu} = 0, \\ \tau_i = \int_{\gamma_i} \sqrt{-g(\dot{\gamma}_i, \dot{\gamma}_i)} d\lambda, \\ \text{RG admissibility, gauge, signature, and boundary constraints.} \end{cases}$$

It maps observable inputs to an observable geometric equivalence class $[g]_{\text{obs}}$, not automatically to a unique latent spacetime ontology.

Theorem 10.2 (Conditional proper-time realization). *Assume that independent clock mechanisms form a universal affine clock; source data satisfy the joint conservation and field-equation constraints; path data identify a stably Lorentzian observable metric class; and one calibration constant predicts held-out admissible paths. Then, on the tested path class,*

$$\chi_{\text{phys}}(\mathfrak{T}[\gamma]) = a \tau_g[\gamma] + b, \quad a > 0.$$

For elapsed durations the origin b cancels.

Proof. The affine-clock theorem supplies one common clock up to positive affine gauge. The identifiable joint inverse problem supplies $[g]_{\text{obs}}$ and hence τ_g on the tested paths. A single calibration fixes a and b ; held-out agreement makes the identification predictive rather than a pathwise refit. $\square$

Non-claim 10.3. The theorem is conditional. Its premises require physical multi-clock, multi-path data and cannot be discharged by synthetic batteries alone.

11 GR bridge no-go results

No-go theorem 11.1 (Source nonuniqueness). *Stress-energy data alone do not generally determine a unique metric. In any linearized inverse problem $Ag = T$ with $\ker A \neq \{0\}$, g and $g + h$ yield the same source for every $h \in \ker A$.*

No-go theorem 11.2 (Vacuum freedom). *$T = 0$ does not imply trivial geometry whenever the homogeneous field equation admits nonzero solutions.*

No-go theorem 11.3 (Path-rank obstruction). *Let local symmetric metric coefficients be represented by $g \in \mathbb{R}^{d(d+1)/2}$ and clock/path observations by $y = Dg$. If D lacks full column rank, then the metric is not identifiable from those observations.*

Proof. Choose nonzero $h \in \ker D$. Then $D(g + h) = Dg$. $\square$

Corollary 11.4 (Conditional joint identification). *Source and path data may jointly identify the observable metric coefficients when the combined design has full rank, stable conditioning, fixed gauge, and stable signature.*

No-go theorem 11.5 (Duration does not determine orientation). *Unsigned quadratic or proper duration is invariant under path reversal and therefore does not select future versus past.*

Proof. For every quadratic form g , $g(-v, -v) = g(v, v)$. $\square$

12 Dimension and ontology no-go results

Theorem 12.1 (Hidden-extension invariance). *Let a latent model M induce $\mathcal{E}$. For any auxiliary space Z , suppose every declared observation on $M \times Z$ factors through the projection to M . Then M and $M \times Z$ induce isomorphic observational quotients and identical natural RG structures.*

Proof. All points in each Z -fibre have the same declared law. Quotienting collapses the fibres and leaves exactly the classes induced by M . $\square$

Corollary 12.2 (Observable rank is not latent dimension). *A four-dimensional Lorentzian observable quotient does not imply a uniquely four-dimensional latent realization.*

No-go theorem 12.3 (Nonfaithful realization). *Without a faithful declared competitor category, equality of GR or RG observable predictions need not imply isomorphism of latent realizations.*

Corollary 12.4 (Primitive-time inference is invalid). *Successful reconstruction of GR proper time does not by itself establish a global time function, preferred foliation, time orientation, unique latent dimension, or primitive temporal ontology.*

13 Final theorem ladder

Layer	Status and exact claim
Experiment-internal quotient	Proved: every equality-of-law invariant factors uniquely through $Q_{\mathcal{E}}$.
Admissibility minimality	Proved componentwise relative to paired stability and gluing properties; global protocol initiality remains open.
Resolved/null sector	Canonical relative to $(\mathcal{E}, \Gamma)$ and the declared regularity class.
Semigroup duration	Proved: $\mathcal{T} = -\log C$ is the continuous additive linearization, up to positive scale.
Extensive scalar clock	Proved in the declared character class: determinant powers, hence $a \text{ tr } \mathcal{T}$.
Affine clock gluing	Proved conditionally from positive affine overlap maps and cocycle closure.
Universal duration object	Open representability theorem; proved only on restricted semigroup strata.
Global memory obstruction	Local triangle statement proved; global H^2 characterization remains open.
GR proper-time bridge	Conditional theorem plus empirical premises; identifiable only with conservation, gauge, boundary, path-rank, stability, and validation gates.
Primitive time or dimension	Not implied; blocked by orientation, hidden-extension, rank, gauge, and nonfaithfulness no-go results.

14 Falsification and validation programme

The framework must be revised if any of the following occurs:

1. a claimed experiment-internal prediction fails quotient invariance;
2. admissible sector rank or conditioning is unstable outside declared tolerances;
3. a claimed transport or clock atlas is path-dependent despite asserted cocycle closure;
4. a normalized continuous additive scalar semigroup duration differs from the logarithmic family;
5. the proposed process-tensor duration fails link-product naturality or cannot separate memory from representation artefacts;
6. a claimed metric is unique under a rank-deficient or gauge-unfixed design;
7. one calibration fails held-out multi-clock, multi-path relativistic tests;
8. a latent ontology claim survives only by excluding same-law hidden extensions without experimental justification.

15 Reproducibility ledger

The theorem search is accompanied by deterministic finite attacks with seed 20260807:

- `rg_admissibility_duration_attack.py/json`: independent witnesses for quotient invariance, rank stability, bounded conditioning, monotonicity, composability, and continuity;

finite candidate attack on logarithmic duration; affine cocycle and triangle-obstruction checks.

- `rg_gr_bridge_attack.py/json`: source/metric kernel freedom, vacuum homogeneous freedom, path-rank nonidentifiability, conditional joint reconstruction, scalarization ambiguity, orientation no-go, and a conservation gate.
- `rg_global_clock_gluings.py`: semigroup additivity and affine atlas consistency.

These computations are implementation checks and counterexample witnesses, not substitutes for the proofs or for physical validation.

16 Non-claims

Non-claim 16.1. RG does not prove that measurement creates physical time.

Non-claim 16.2. RG does not refute GR proper time. It places proper time as a conditional physical representation of admissible observable duration and path structure.

Non-claim 16.3. The word “canonical” applies unconditionally to quotient factorization relative to the declared experiment. Enriched metrics, transports, clocks, and realizations are canonical only relative to their characterization hypotheses and categories.

Non-claim 16.4. Reconstructibility from observation is not a metaphysical proof that observation is more fundamental than spacetime.

17 Conclusion

The stable foundation of Resolution Geometry is not a preferred metric but the observable quotient and its resolved/null split. A declared experiment removes distinctions that cannot affect its complete observation laws; an admissibility protocol retains the stable and composable observable sector. On the positive commuting semigroup stratum, process composition then determines logarithmic operator duration. Scalar clocks are representations of that duration, and compatible physical clock mechanisms glue as affine charts of one common clock object.

General relativity enters neither as a primitive assumption of the abstract construction nor as something derived without physical input. It is a constrained realization of admissible clock-path and source observables. Under conservation, gauge, boundary, rank, signature, and validation conditions, the bridge can reconstruct an observable Lorentzian metric and identify a universal clock with GR proper time on a tested path class. Yet the bridge is non-faithful at the ontological level: it does not force global time, time orientation, a preferred foliation, unique latent dimension, or primitive temporal ontology.

The strongest defensible synthesis is therefore

observable duration is reconstructible from admissible process composition,
 GR proper time may be its validated physical representation,
 but successful representation does not uniquely determine temporal ontology.

The remaining mathematical closure is a representability theorem for universal duration, including process-tensor memory. The remaining physical closure is a held-out cross-clock, multi-path GR realization test.

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